Denoising Diffusion Implicit Models: Accelerated Sampling and Quality Trade-offs
EEE 598: Generative AI Theory and Practice — Project Proposal

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Reference Paper
Song, J., Meng, C., & Ermon, S. (2021). *Denoising Diffusion Implicit Models*. In *International Conference on Learning Representations (ICLR 2021).* arXiv:2010.02502.

Proposed Work
This project studies whether diffusion models can generate high-quality images with far fewer sampling steps than standard DDPM methods. Our work is grounded in the DDIM paper by Song, Meng, and Ermon, which shows that sampling can be made much faster by replacing the original stochastic reverse process with a non-Markovian implicit process, without changing the training objective. We will focus mainly on the practical side of the paper, while also explaining the core theoretical idea behind DDIM sampling. Using the authors' official implementation and a pretrained CIFAR-10 DDPM checkpoint, we will reproduce the paper's central comparison between DDPM-style and DDIM-style sampling at different step counts. We will evaluate image quality with FID and compare generation speed using inference time. In addition, we will test intermediate values of the stochasticity parameter eta to see how partial stochasticity affects performance. Our main goal is to identify a realistic speed-quality trade-off and determine when DDIM provides the best balance between efficiency and output quality. Stretch goals include comparing timestep schedules and briefly connecting DDIM-style sampling to later systems such as latent diffusion models.

Expected Results / Datasets
We'll use CIFAR-10 and a pretrained DDPM checkpoint to compare DDPM and DDIM sampling across step counts. We expect DDIM to preserve image quality better than DDPM at low numbers of steps, while also reducing inference time substantially; results will be reported through FID, runtime comparisons, and generated image samples.

References

[1] Song, J., Meng, C., & Ermon, S. (2021). *Denoising Diffusion Implicit Models*. In *International Conference on Learning Representations (ICLR 2021).* arXiv:2010.02502.

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[3] Heusel, M., Ramsauer, H., Unterthiner, T., Nessler, B., & Hochreiter, S. (2017). *GANs Trained by a Two Time-Scale Update Rule Converge to a Local Nash Equilibrium*. In *Advances in Neural Information Processing Systems (NeurIPS 2017).* arXiv:1706.08500.

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[5] Rombach, R., Blattmann, A., Lorber, D., Esser, P., & Ommer, B. (2022). *High-Resolution Image Synthesis with Latent Diffusion Models*. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2022).* arXiv:2112.10752.

Supporting Resource
Ermon Group. *Official DDIM implementation*. GitHub repository.